UNIVERSITY OF PLYMOUTH

BSC (HONS) COMPUTER SCIENCE (SOFTWARE ENGINEERING)

MAL3018 COMPUTING PROJECT

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Date	**:**	14 September 2026

IsyaratKu – Sign Language Recognition System with MediaPipe

Keywords: Bahasa Isyarat Malaysia (BIM), Sign Language Recognition, MediaPipe, Gesture Recognition, Assistive Technology

Links:

1. **GitHub** : https://github.com/Hoo-Sulyn/MAL3018_IsyaratKu
2. **PebblePad** : https://v3.pebblepad.co.uk/spa/#/public/mtw79Z898W44cr9scsskMg73GW

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1.0 Introduction

1.1 Background

Bahasa Isyarat Malaysia (BIM) is the official sign language of the Deaf community in Malaysia, legally recognised under the Persons with Disabilities Act 2008. However, BIM has yet to be named Malaysia's national sign language, and compliance with the Act's provisions remains inconsistent in practice.

This gap is compounded by a severe shortage of interpreters. According to Wan Zuraidah Abu, president of the Malaysian Association of Sign Language Interpreters (MyAsli), only around 60 active interpreters serve over 44,000 deaf Malaysians nationwide, and in some states just one or two interpreters cover up to 3,000 deaf people (The Star, 2025). This scarcity widens the communication gap between Deaf and hearing Malaysians, directly affecting BIM signers seeking to communicate in everyday settings, and hearing individuals — such as educators, healthcare workers, or family members — who lack the means to understand BIM.

Accurate hand-gesture tracking for sign language recognition has historically required specialised hardware such as multi-camera rigs or coloured gloves. The introduction of frameworks such as MediaPipe has substantially lowered this barrier, enabling real-time hand-landmark extraction from a single standard camera, efficient enough to run on consumer mobile devices. This shift makes it newly feasible to develop functional gesture recognition systems without specialised equipment, opening the door for under-resourced languages like BIM to receive dedicated recognition tools for the first time.

This project aligns with SDG 10 (Reduced Inequalities, Target 10.2), SDG 4 (Quality Education), and SDG 9 (Industry, Innovation and Infrastructure), by contributing toward inclusive communication and foundational technological infrastructure for a currently under-resourced language (United Nations, 2015).

1.2 Problem Statement

Despite BIM being the official sign language of Malaysia's Deaf Community, no existing sign language recognition system currently supports it. This gap is consistent with prior observations that BIM-related datasets remain small and largely limited to fingerspelling and a small number of dynamic words (Chong et al., 2026), and is further confirmed by the systematic comparison of existing recognition-direction systems presented in *4.0 Literature Review*, none of which currently support BIM. This project addresses that gap by developing a static BIM gesture recognition system capable of recognising fingerspelling and common vocabulary, and composing them sequentially into words and short phrases.

1.3 Target Users

The target users of IsyaratKu are BIM signers and hearing individuals who do not know BIM, interacting with each other through a single Android device. BIM signers use the application by signing static gestures, such as fingerspelling or common vocabulary, in front of the camera. Hearing users read the recognised text and can reply by typing a message through the note feature. Typical hearing users include family members, educators, and healthcare or service staff who need to communicate with a Deaf person in everyday situations where an interpreter is not available. The application is designed for ordinary Android smartphone users, requiring no specialist equipment or prior technical knowledge.

2.0 Objectives

The main objectives of this project are:

1. To design and develop a static BIM gesture recognition pipeline using MediaPipe hand-landmark extraction combined with a machine learning classifier.
2. To evaluate and compare classifier approaches (e.g. SVM/Random Forest versus a neural network) for recognition accuracy.
3. To extend recognition beyond fingerspelling into a set of common BIM vocabulary words.
4. To implement the trained model into a working Android application with a sequential capture mechanism that composes recognised gestures into words or short phrases.

3.0 Scope of the Project

In-Scope Features:

- Recognition of static BIM gestures, including sequential composition of individual gestures into words or short phrases (e.g. spelling letter-by-letter or stringing vocabulary signs together).
- Fingerspelling recognition, covering the alphabet (A–Z) and numerals (0–9).
- Recognition of a small, defined set of common everyday BIM vocabulary.
- Deployment as an Android application featuring a live camera-based recognition demonstration.
- A supplementary two-way text note feature, enabling a hearing user to input a written message for the Deaf user to read.

Out of Scope:

- Dynamic or continuous sentence-level sign translation, which requires temporal sequence modelling and substantially larger annotated datasets beyond the scope and timeline of this project.
- Comprehensive BIM vocabulary coverage, as sufficiently large datasets are not currently publicly available.
- Recognition of multi-hand or two-handed complex signs, as the datasets employed are predominantly single-hand.
- Real-time translation of recognised output into spoken audio, which would require a separate text-to-speech component outside this project's scope.
- Grammatical or sentence-structure correction of composed output, as BIM grammar differs structurally from written Malay or English.

4.0 Literature Review

Sign language technology broadly splits into two directions, generation (converting spoken or written language into sign output, usually via an animated or AI-generation avatar) and recognition (converting a signer's gestures into text or speech output). This project addresses the recognition direction.



(*Figure 4.1: Google DeepMind's SL2T. Source: Sterling, 2026*)

Among recognition-direction systems, Google DeepMind's SL2T is the most technically advanced, allowing users to sign continuously while the system translates their gestures into text or sentences in real time, powering sign-to-text dictation in Gboard and Live Transcribe on the Pixel 11. At present, it supports ASL-to-English translation only, with no coverage of BIM or other under-resourced sign languages (Google DeepMind, 2026).



(*Figure 4.2: PopSign AI. Source: Popsign.org, 2023*)

PopSignAI takes a different approach, using static, landmark-based recognition within an educational game rather than as a communication tool. Developed through a collaboration between Georgia Institute of Technology and RIT/NTID's Center on Access Technology, it teaches learners over 500 ASL signs, drawing its sign selection from the MacArthur-Bates Communicative Development Inventories – a framework originally designed to track early language acquisition in children (PopSign, 2023).



(*Figure 4.3: SignAll. Source: CB Insights, 2016*)

SignAll focuses on bridging communication gaps between Deaf and hearing individuals through sign language translation technology (CB Insights, 2016). One of its products, SignAll Chat, translates ASL into English text in real time by recognising isolated signs and composing them into sentences using rule-based grammar, enabling spontaneous, real-time conversation. SignAll originally relied on physical gloves and marker-based tracking, later moving toward glove-free, camera-based tracking, reflecting the broader industry shift toward simpler, single-camera input (Future of Interface, 2023).



(*Figure 4.4: Sign-Speak. Source: Sign-Speak, 2026*)

Sign-Speak provides automated ASL interpretation, captioning, and translation services. One of its products, SignCaption, automatically captions ASL videos: users upload or create ASL video content, and the system generates and syncs captions using continuous sign recognition. Unlike SignAll and PopSign, Sign-Speak does not publicly disclose its underlying recognition technique, limiting its usefulness as a technical benchmark (Sign-Speak, 2026).

Table below summarises and compares the technical characteristics of the systems discussed above alongside this project.

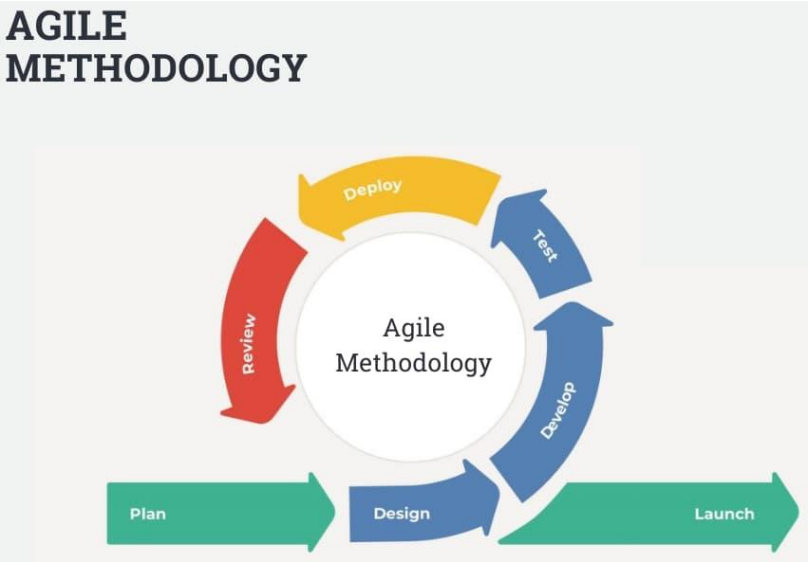
Feature	SL2T	PopSign	SignAll	Sign-Speak	This Project – IsyaratKu
Landmark-based recognition	✓	✓	✓	?	✓
Static gesture recognition	×	✓	✓	×	✓
Continuous/dynamic recognition	✓	×	×	✓	×
Sequential word/phrase composition	✓	×	✓	✓	✓
Publicly disclosed technique	✓	✓	✓	×	✓
Mobile app deployment	✓	✓	×	×	✓
Real-time translation	✓	✓	✓	×	✓
Supports BIM	×	×	×	×	✓

(*Table 4.1: Comparison of features between systems discussed above alongside this project*)

Across the systems reviewed, recognition-direction technology has been developed almost exclusively for ASL. As Table 4.1 illustrates, no existing system currently supports BIM. This project isn't a smaller competitor to systems like SL2T, but a scoped contribution to a language that recognition technology hasn't reached at any scale, combining fingerspelling and vocabulary recognition with sequential capture in a deployed Android prototype for BIM.

5.0 Methodology

5.1 Project Approach

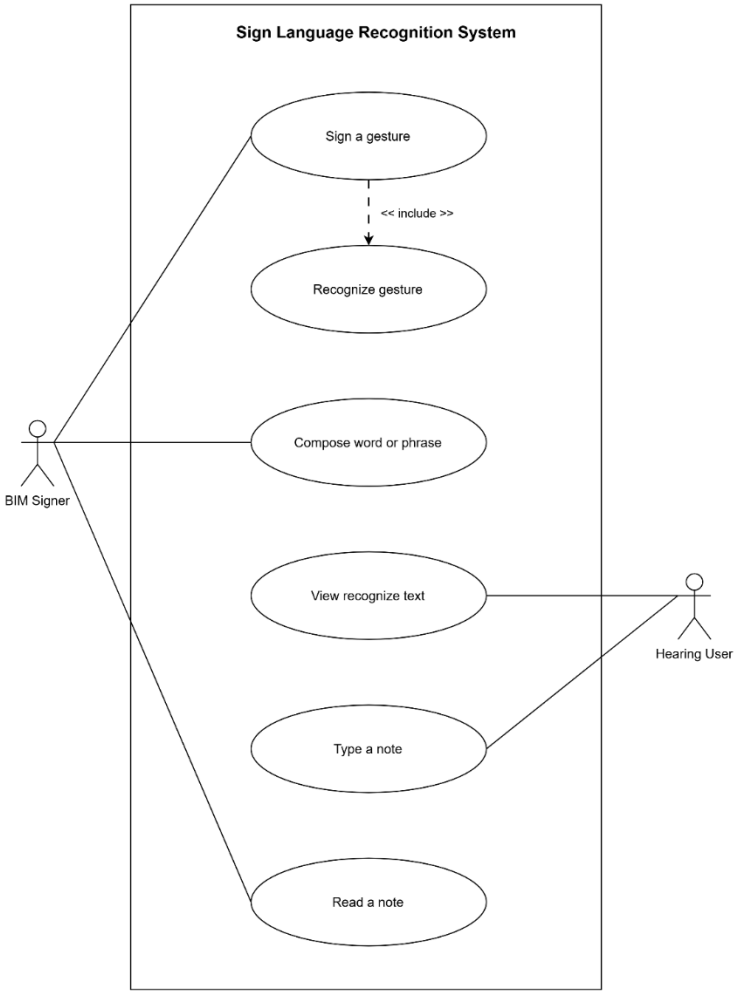


(*Figure 5.1: Agile Methodology. Source: INTERQUALITY, 2024*)

This project follows an Agile methodology, in which development is divided into short cycles of about two weeks instead of being planned and built in a single pass. Each cycle produces a working increment, such as the landmark extraction pipeline, a trained classifier, or an app module. The supervisor reviews it, and the feedback shapes the next cycle. This suits the project because model training involves repeated experimentation, and the results of one cycle often change what needs to happen in the next (Institute of Project Management, 2025).

5.2 UML Diagrams

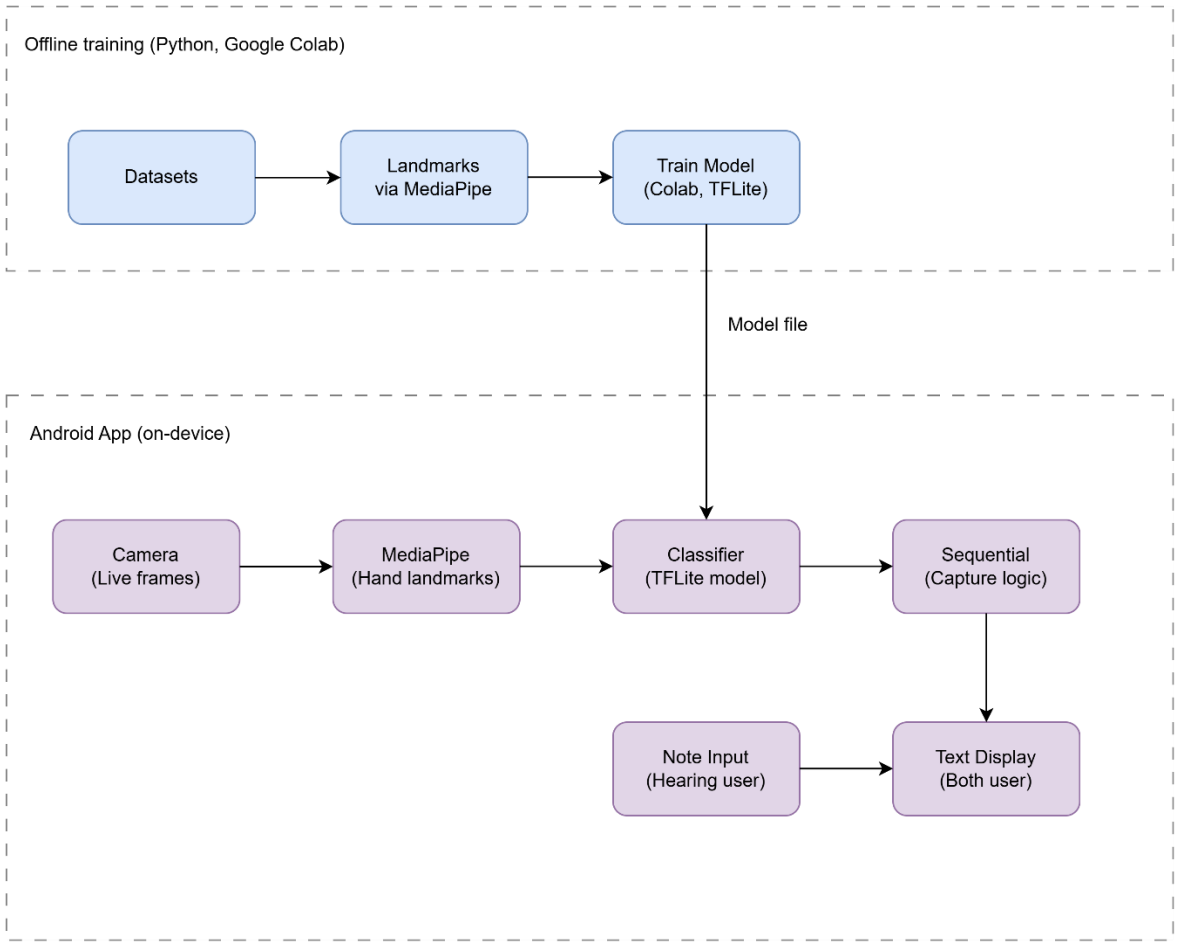
5.2.1 Use Case Diagram



(*Figure 5.2.1: Use Case Diagram*)

Figure 5.1 shows the two types of user and the functions each can perform within the system. The BIM signer signs a gesture in front of the camera, which includes the system recognising that gesture, composes recognised gestures into a word or phrase, and reads notes written by the hearing user. The hearing user views the recognised text and types a note in reply. Both users interact with the application on a single Android device, so the diagram describes two roles within one system, not two separate systems.

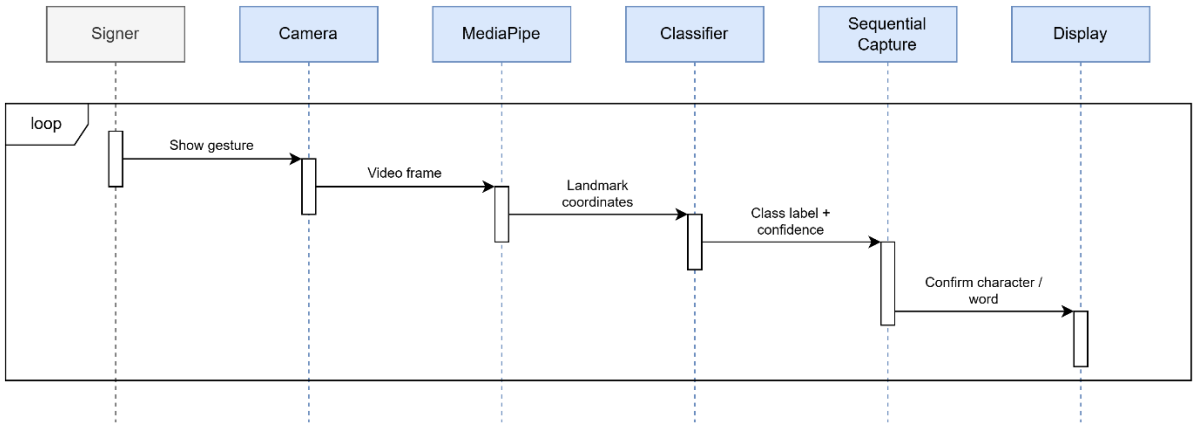
5.2.2 System Architecture Diagram



(*Figure 5.2.2: System Architecture Diagram*)

Figure 5.2 shows the system in two stages. In the offline training stage, carried out in Python on Google Colab, images from the datasets are passed through MediaPipe to extract hand landmarks, which are used to train the classifier. The trained model is converted to TensorFlow Lite and exported as a model file. In the on-device stage, the Android application captures live frames from the camera, extracts hand landmarks with MediaPipe, and passes them to the classifier, which loads the exported model. The sequential capture logic then confirms predicted signs and appends them to the text shown on the display, which is shared by both users. Text typed by the hearing user through the note input appears on the same display. Because all processing takes place on the device, the application needs no backend server.

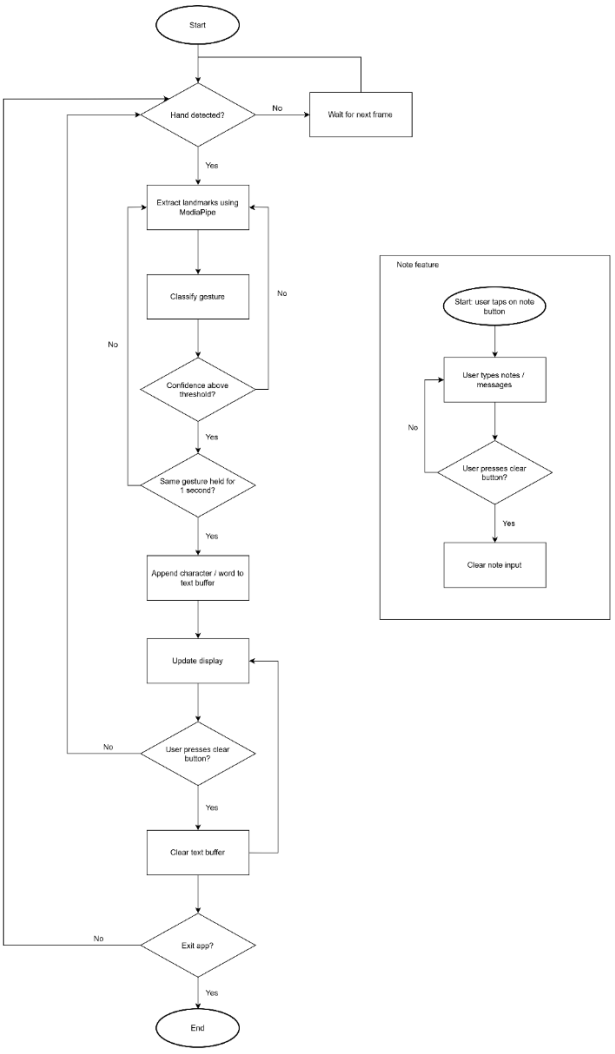
5.2.3 Sequence Diagram



(*Figure 5.2.3: Sequence Diagram*)

Figure 5.3 shows the order of interactions for each gesture. The signer shows a gesture to the camera, which passes a video frame to MediaPipe. MediaPipe sends the landmark coordinates of the hand to the classifier, which returns a class label with a confidence score. The sequential capture component receives this result and, once the sign is confirmed, passes the character or word to the display. The loop frame indicates that this cycle repeats for every gesture while the application is in use.

5.2.4 Flowchart



(*Figure 5.2.4: Flowchart*)

Figure 5.4 describes the logic that runs for each camera frame. The system first checks whether a hand is detected, and if not, waits for the next frame. When a hand is present, landmarks are extracted and the gesture is classified. A prediction is accepted only if its confidence is above a threshold and the same gesture has been held for about one second. This prevents flickering predictions while the hand moves between signs. Accepted characters or words are appended to the text buffer and the display is updated. The user can clear the text buffer with a clear button, and the loop continues until the user exits the application. The separate note feature shows that typing and clearing a note operates independently of the recognition loop. The confidence threshold and hold duration are initial values and will be tuned during testing.

6.0 Expected Deliverables

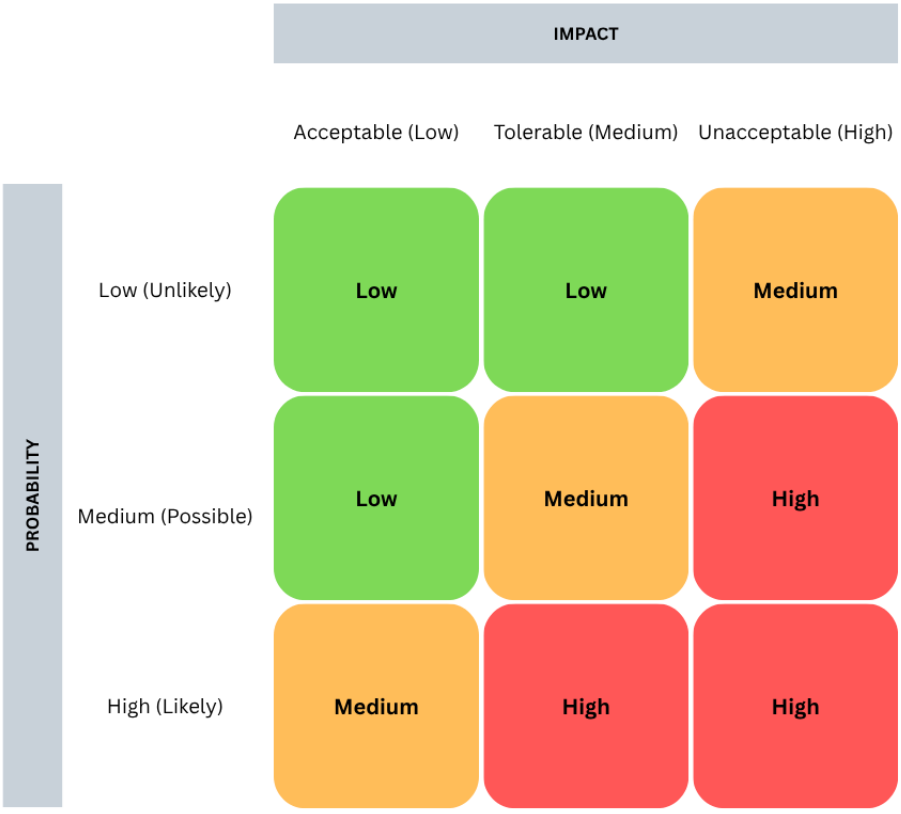
IsyaratKu, a mobile-based sign language recognition application, is expected to deliver direct benefits to both of its user groups. For Deaf users, the application offers a lightweight alternative for everyday communication when an interpreter is not available, recognising and composing signed BIM gestures sequentially into text as the user signs. For hearing users, IsyaratKu not only enables understanding of what a BIM signer is communicating, but also allows them to type a message using the note feature, enabling two-way interaction between both users.

7.0 Risk Plan

No.	Risk	Likelihood	Impact	Mitigation
1	The vocabulary dataset is too small or too inconsistent to train a reliable model.	Medium	Medium	Treat vocabulary as an extension. The core deliverable relies on the larger fingerspelling dataset. If vocabulary underperforms, scope the demo to fingerspelling only.
2	Dataset signs don't match standard BIM or are mislabelled.	Medium	High	Check a sample of images against BIM reference material early, and ask a BIM signer or interpreter to verify if possible.
3	Model accuracy is too low for a convincing demo	Medium	High	Start with a simple baseline, compare several classifiers, and use data augmentation. Keep a fallback of a smaller sign set with higher accuracy.
4	MediaPipe landmark extraction fails under poor lighting, cluttered backgrounds or unusual hand angles	Medium	Medium	Test on varied footage early and add lighting and background guidance to the app UI.
5	The trained model can't be converted to TensorFlow Lite, or runs too slowly on a phone	Medium	High	Build the neural network in TensorFlow/Keras so it converts cleanly, and test on-device performance early, not at the end.

6	Scope creep (extra features, continuous recognition)	Medium	High	Follow the documented scope. Record new ideas as future work, and accept only trivial additions.
7	Difficulty finding test users, especially Deaf community members	High	Medium	Contact deaf associations early, and use trained volunteers as a fallback.

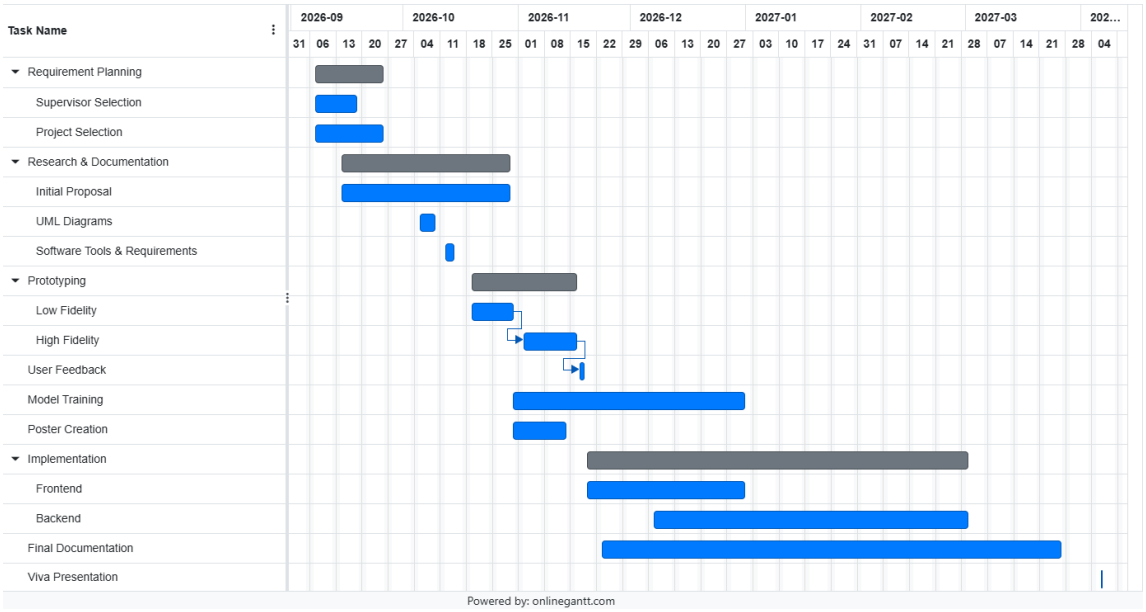
(*Table 7.1: Risk Plan Table*)



(*Figure 7.1: Risk Assessment Matrix*)

8.0 Project Timeline

Below is an initial Gantt Chart depicting the project's progress:



(*Figure 7.1: Gantt Chart of IsyaratKu Project*)

9.0 Resources Required

- **Hardware Requirements**

  Below are the hardware requirements needed for the project:

Category	System Name
Processor	13th Gen Intel(R) Core(TM) i5-13420H (2.10 GHz)
RAM	16.00 GB
System Type	64-bit operating system, x64-based processor

(*Table 8.1: Hardware Requirements*)

- **Software Requirements**

  Below are the software requirements needed for this project:

Category	System Name
Code Editor / IDE	Android Studio
Version Control	GitHub
Frontend Development	XML
Machine Learning / Model Training	Python, MediaPipe, scikit-learn, TensorFlow/Keras
Model Deployment	TensorFlow Lite
Dataset Sources	Kaggle
Prototyping / UI Design	Figma
DevOps & Deployment	GitHub

(*Table 8.2: Software Requirements*)

10.0 Expected Outcome and Significance

This project is expected to produce a working Android prototype that recognises static BIM gestures through the device's live camera and composes them sequentially into words and short phrases. The underlying classifier is expected to achieve at least 85% accuracy on held-out test data. A comparison of classifier approaches, namely SVM/Random Forest against a neural network, is expected to identify which performs best for landmark-based BIM recognition.

IsyaratKu is expected to benefit both of its user groups. For Deaf BIM signers, it offers a lightweight alternative for everyday communication when an interpreter is not available. For hearing users, it enables them to understand what a signer is communicating and to reply by typing a message through the note feature, supporting two-way interaction between both parties.

Academically, the project provides a documented baseline for landmark-based BIM recognition, a language with very little prior work in this area, which future research can build on. In societal terms, the project supports SDG 10 (Reduced Inequalities) by promoting the inclusion of persons with disabilities, with supporting contributions to SDG 4 (Quality Education) and SDG 9 (Industry, Innovation and Infrastructure) (United Nations, 2015).

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Appendix

Appendix A: UML Diagram Links

Use Case Diagram	:	https://drive.google.com/file/d/11NE5FruewFEcdLr0Gx9CvVpk8dN45nk0/view?usp=sharing
System Architecture Diagram	:	https://drive.google.com/file/d/1BVwMllvxvx5ZoYOpTDCLxWGZZl_xjJkk/view?usp=sharing
Sequence Diagram	:	https://drive.google.com/file/d/1Rrzv1wrHjlXrFaNya8OGCZJh6BNCHpAa/view?usp=sharing
Flowchart	:	https://drive.google.com/file/d/1YahvcoLEvVko4YdXQqo-WFNN9SSrp-Gz/view?usp=sharing

Appendix B: Gantt Chart Link

Google Drive : https://drive.google.com/file/d/1-ITxI_rnci74bxeowjVd5mX3xoDs9WfE/view?usp=sharing